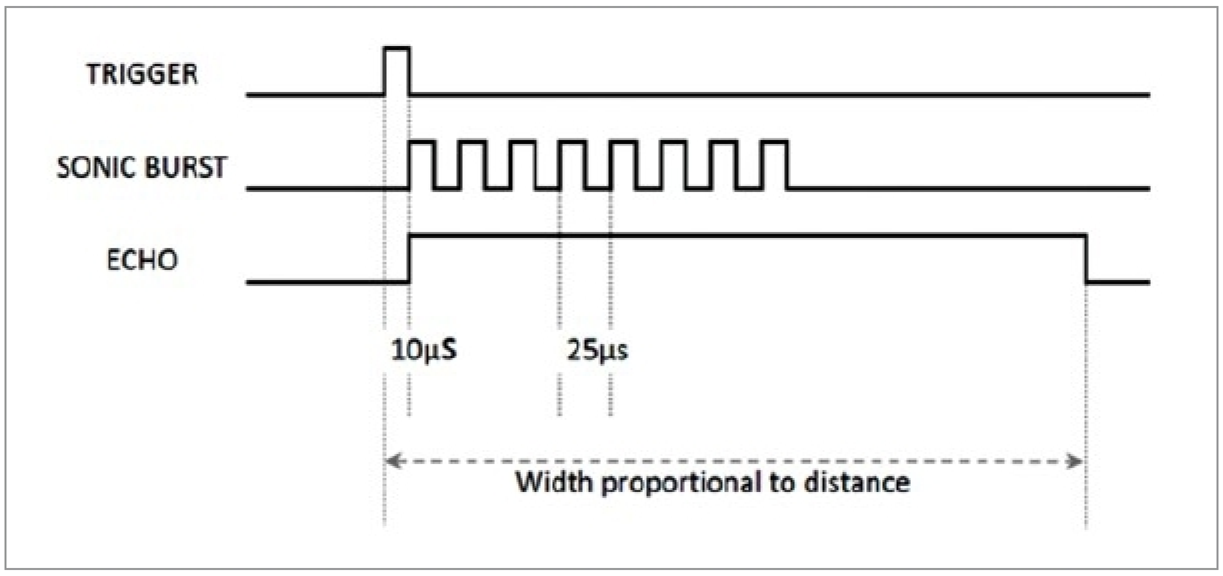

Fig. 4: Timing diagram of ultrasonic sensor

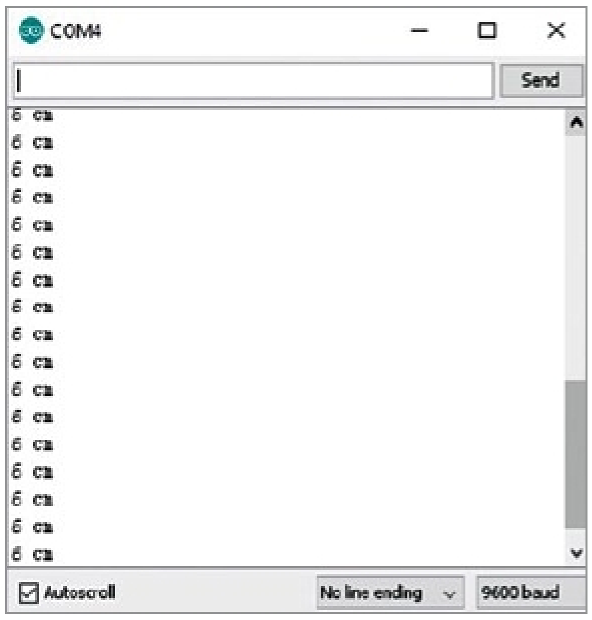

Fig. 5: Sensor output from serial monitor

and paste it in under Documents→Arduino→Libraries folder.

Construction and testing

Assemble the components as per the circuit diagram given in Fig. 3. Connect the USB cable with Arduino Uno. Open Arduino IDE and go to File→Examples→Ultrasonic→ButtonSerial, and open the code.

Compile and upload this code to Arduino Uno. Open Serial Monitor under Tools of Arduino IDE to see the sensor output (distance between sensor and object). Make sure the baud rate is 9600. Note that alignment of the ultrasonic module with the object plays a major role in getting accurate sensor output.

Keep an object near the sensor. When switch S1 is closed, the sensor transmits ultrasonic waves. The distance between the object and sensor is displayed on the serial monitor, as shown in Fig. 5. EFY

Arnav Bansal is currently employed as R&D engineer in an automation company. He primarily works on custom Internet of Things (IoT) devices